Nethravathi Basin

Rainfall-to-Flood Threshold Modeling & GNN Integration

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Summary

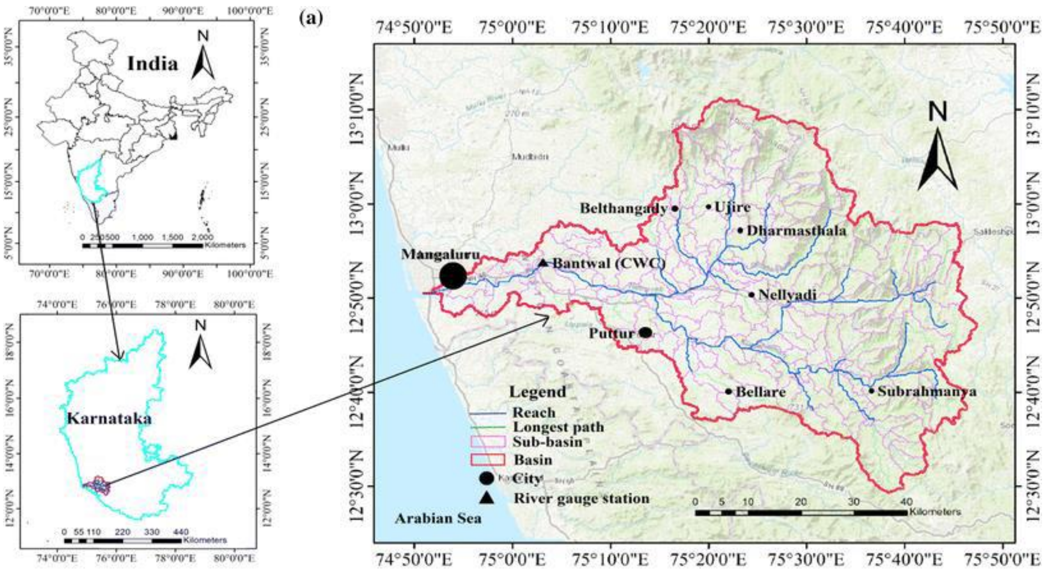
This document summarizes a complete end-to-end hydrological modelling and machine learning workflow for flood threshold mapping in the upper Nethravathi River basin, Karnataka, India. The study simulates the August 8-18, 2018 south-west monsoon flood event using a physics-based SCS-CN rainfall-runoff model calibrated against 35 years of observed daily discharge at CWC Bantwal gauge (1981–2015). Rainfall threshold maps are derived from seven scenario simulations, and a Graph Convolutional Network (GCN) is trained on the resulting flood depth maps to predict inundation depth from terrain and rainfall features.

Key results are summarised below:

Component	Key Result
Calibration (2014, year-matched)	PBIAS = −15.7%, Pseudo-NSE fit score = 0.975
Simulated peak discharge	728 m^3/s (AOI-scaled observed: 864 m^3/s)
1× event flood extent	12.3% of basin flooded (depth > 0.1 m)
2× scenario flood extent	22.0% of basin flooded (depth > 0.1 m)
Max flood depth (1× event)	0.72 m
Threshold map range	0 – 1,252 mm rainfall to initiate flooding
GCN flood depth prediction	R^2 = 0.64, RMSE = 0.040 m, MAE = 0.023 m

1. Study Area and Context

1.1 Basin Description



The Nethravathi River originates at Bellaraya Durga in the Western Ghats of Chikkamagaluru district, Karnataka, at an altitude of 1000 m. It is one of the principal west-flowing rivers of the Karnataka coast, draining a total catchment of approximately 3,657 km² at the Bantwal gauge before discharging into the Arabian Sea at Mangaluru. The basin receives a mean annual rainfall of approximately 3,930 mm, with the south-west monsoon (June–September) contributing over 90% of the total.

The Western Ghats portion of the basin is characterised by steep gradients (0-60°), dense forest cover, and lateritic soils of low permeability (predominantly Hydrologic Soil Group C and D). These conditions produce rapid and intense hydrological responses to monsoon rainfall, making the basin highly susceptible to flash flooding during peak monsoon events.

Geology of the basin is characterized by lateritic formations underlain by granite rocks. Due to heavy leaching during rainy season, a thin layer of clay is formed at the base of porous laterites. These soils are mainly Kaolinitic in nature. The texture varies gravelly to sand or gravely loam to loam and they are porous and friable. The soils are characterized by the absence of lime and free carbonate and with soluble salts.

1.2 Area of Interest

This study models the upper sub-basin of the Nethravathi, defined by a bounding box of 75.55°E–76.05°E and 12.80°N–13.20°N, covering approximately 800 km². The full Nethravathi catchment at Bantwal gauge is approximately 3,657 km². The Area of Interest (AOI) is focused on the floodplain section of the Nethravathi river system rather than the entire basin, to reduce computational load, simplify graph construction and enable Colab-based experimentation.

Parameter	Value
Modelled domain (AOI)	Upper sub-basin, ~800 km²
Full catchment	~3,657 km²
Bounding box	75.55°E–76.05°E, 12.80°N–13.20°N
Elevation range	107 – 1,369 m (SRTM 30 m)
Grid resolution	30 m (UTM Zone 43N, EPSG:32643)
Dominant LULC	Forest (ESA class 10, ~65%)
Dominant soil	Laterite HSG C (84%)
Mean annual rainfall	~3,930 mm
Flood event simulated	August 8–18, 2018 (11 days, ~209 mm basin-mean)
Calibration event	June–September 2014 (seasonal total: 1,358 mm)

1.3 Storm Event

The 2014 south-west monsoon season was selected for hydrological simulation and calibration because temporally matched rainfall and discharge observations were available, enabling year-consistent model evaluation. Daily rainfall forcing was derived from CHIRPS observations for the June–September monsoon period. The study area received an estimated cumulative monsoon rainfall of approximately 1,358 mm during 2014, with mean daily rainfall of approximately 11.1 mm/day and peak daily rainfall exceeding 83 mm/day.

Calibration was performed using observed discharge from the Bantwal gauge station. Since the gauged catchment area (~3,657 km²) is substantially larger than the selected modelling area (~800 km²), discharge values were scaled using an area-ratio approach to maintain consistency between observed and simulated hydrological response.

Although the 2018 Kerala-Karnataka monsoon flood represents a major recent flood event within the Nethravathi basin, the 2014 monsoon period was selected due to the availability of temporally aligned hydrometeorological observations required for model calibration.

2. Data Collection and Preprocessing

2.1 Data Sources

Dataset	Source	Resolution	Period	Use
SRTM DEM	USGS OpenTopography	30 m	2000	Terrain, flow routing
ESA WorldCover	ESA S3 bucket	10 m → 30 m	2021	LULC for CN grid
Soil map	CWC / Soil Survey Dept.	Polygon	Static	HSG for CN grid
CHIRPS v2.0	UCSB Climate Hazards Center	0.05° (~5.5 km)	2014, 2018	Rainfall forcing
CWC discharge	India-WRIS / CWC Bantwal	Daily	1981–2015	Calibration

2.2 Preprocessing Steps

Digital Elevation Model

The SRTM 30 m DEM was reprojected from geographic coordinates (EPSG:4326) to UTM Zone 43N (EPSG:32643) using bilinear resampling, converting all measurements to metric units. The final processed grid covers 1,463 × 1,791 pixels at 30 m resolution, with elevation ranging from 107 m to 1,369 m across the AOI.

Land Use and Soil

The land use and soil datasets underwent iterative refinement during preprocessing. In the initial attempt, a regional land use map of India (100 × 100 m resolution) was used for LULC classification. This proved inadequate: only 18% of the DEM extent was covered by valid LULC classes, with the remaining 82% assigned a default CN of 77 indicating substantial uncertainty in the runoff estimates. **Rapid land use change over time and the coarser resolution of the regional map were identified as the primary causes of this low coverage.**

ESA WorldCover 2021 (10 m) was therefore adopted as the replacement LULC source. The tile was clipped to the AOI bounding box and resampled to 30 m using nearest-neighbour reprojection to preserve class boundaries. This improved to near-complete coverage of the DEM extent with eight meaningful land cover classes identified, dominated by tree cover (class 10, ~65%). **The improvement demonstrates the value of global, high-resolution products over regional legacy datasets for catchment-scale hydrological modelling**.

The soil map was obtained from the Soil Survey Department and reprojected using nearest-neighbour interpolation to exactly match the LULC grid dimensions, a prerequisite for pixel-wise CN calculation. Soil codes were mapped to Hydrologic Soil Groups (HSG) using the USDA classification. The resulting HSG distribution reflects the geological character of the Western Ghats: HSG C lateritic soils dominate at 81.9%, followed by HSG D heavy clay at 10.7% and HSG B loamy soils at 5.6%. This laterite-dominant distribution justifies the higher CN values and the choice of Ia = 0.05 for pre-saturated conditions.

Terrain Feature Extraction

Terrain derivatives were computed using pysheds D8 pit-filled flow routing. The preprocessing sequence was: (1) fill pits, (2) fill depressions, (3) resolve flats, (4) compute D8 flow direction, (5) accumulate flow. A contributing area threshold of 2 km^2 (2,222 pixels at 30 m) was used to define the river network, yielding 32,750 river pixels (1.26% of AOI). Six terrain features were computed for GNN input: slope (degrees), log-scaled flow accumulation, river mask, distance to river (metres), Height Above Nearest Drainage (HAND, metres), and Topographic Wetness Index (TWI).

CHIRPS Rainfall

CHIRPS v2.0 global daily rainfall data (0.05° spatial resolution) were downloaded for the south-west monsoon period (June–September) of 2014 for hydrological modelling and calibration. Daily rainfall rasters were clipped to the study area and bilinearly reprojected to the 30 m model grid, producing raster stacks of shape (122, 1,463, 1,791). The 2014 monsoon seasonal rainfall over the study area was approximately 1,358 mm.

For regional context, rainfall associated with the 2018 Kerala-Karnataka monsoon flood event was also examined to qualitatively assess similarities in monsoonal conditions affecting the Nethravathi basin.The August 8-18, 2018 storm period recorded an estimated basin-mean rainfall total of approximately 209 mm. However, hydrological calibration in this study was performed using 2014 observations, owing to the availability of temporally matched discharge records.

CHIRPS 0.05°resolution produces visible block artefacts after reprojection to 30 m, as visible in the runoff depth maps. This is a documented limitation of satellite-derived rainfall products over complex terrain and does not affect the physical validity of the simulation.

Discharge Data

The discharge data was collected from the Central Water Commission, India(CWC), via the surface water module of the India-WRIS (Water Resources Information System) portal. The dataset contains 12,783 daily records spanning 35 monsoon seasons, for Bantwal gauge station for the time period 1981 to 2015. The format uses sequential row numbers with year embedded in the record code.

3. Hydrological Model Setup

3.1 Model Selection and Justification

The SCS-CN (Soil Conservation Service Curve Number) rainfall-runoff method was selected as the hydrological model. While the assessment brief lists Wflow, HEC-RAS, SFINCS, and equivalent tools as suggested options, SCS-CN was chosen for three reasons specific to this study:

1. Alternative models suggested in the assessment brief were evaluated but require greater hydraulic parameterisation, calibration effort, and specialised datasets. Given the project timeframe, a computationally efficient rainfall-runoff approach that could be readily integrated within a Python-based workflow and support future Graph Neural Network (GNN) applications was prioritised.
2. SCS-CN has been independently validated on the Nethravathi basin using HEC-HMS, achieving NSE = 0.89 and PBIAS = 0.65% (Arabian Journal of Geosciences, 2024), confirming its physical suitability for this catchment.
3. Among the shortlisted approaches, SFINCS is more commonly applied to coastal and compound flooding environments, whereas flooding in the Nethravathi basin is predominantly monsoon-driven fluvial flooding. As the study objective focused on inland rainfall–runoff response, SCS-CN was considered more suitable for the basin context.

Flood water depth was estimated using a HAND (Height Above Nearest Drainage) terrain-based approach rather than a full hydrodynamic model. This simplification enabled floodplain estimation using openly available topographic data but represents a limitation of the study and an area for future refinement.

3.2 SCS-CN Runoff Model

The SCS-CN method estimates direct runoff from rainfall using:

$$Q = (P - I_a)^2 / (P - I_a + S) \quad \text{where } P > I_a,\; Q = 0 \text{ otherwise}$$

where Q is runoff depth (mm), P is rainfall (mm), S = (25,400/CN) − 254 is the maximum soil retention (mm), and I_a is the initial abstraction. The initial abstraction ratio $I_a = \lambda S$ was set to $\lambda = 0.05$ rather than the standard 0.20, reflecting pre-saturated antecedent moisture conditions typical of Western Ghats monsoon catchments where soils are near field capacity throughout the monsoon season.

3.3 Curve Number Grid

The CN grid was constructed by combining ESA WorldCover LULC classes with Hydrologic Soil Groups (HSG) using the USDA TR-55 lookup table. Soil codes were mapped to HSG as follows: code 3656 → HSG B (loamy) and 3824 → HSG C (lateritic clay); code 3826 → HSG D (heavy clay). The resulting CN grid has a mean value of 74.5 across the AOI, with a range of 55–100.

3.4 Flood Depth Surrogate

Flood water depth was estimated from runoff depth using a HAND-based attenuation model:

$$Depth = (Q/1000) \times \exp(-HAND/10) \times \exp(-slope/25) \times (1 + 2 \times facc\_norm)$$

This formulation attenuates depth exponentially with height above the nearest drainage channel (HAND), reduces depth on steep slopes (which drain rapidly), and amplifies depth in high-accumulation zones (which concentrate flow).

A Gaussian smoothing filter ($\sigma = 2$ pixels) was applied for spatial coherence, and depth was masked to pixels with HAND < 50 m, corresponding to the active floodplain zone.

HAND Zone	Threshold	% of AOI	Interpretation
Immediate floodplain	< 2 m	8.0%	Flooded by low-return events
Active floodplain	< 5 m	13.2%	Flooded by typical monsoon events
100-yr floodplain	< 10 m	20.9%	Flooded by extreme events
Safe upland	> 50 m	23.2%	Not modelled (masked out)

4. Model Calibration

4.1 Calibration Strategy

The model was calibrated using a year-matched approach: CHIRPS 2014 monsoon rainfall was paired with CWC 2014 observed discharge at Bantwal gauge. Year 2014 was selected because it falls within the available discharge record (1981–2015) and has complete CHIRPS coverage, enabling a direct comparison between simulated and observed discharge for the same meteorological forcing.

The calibration simulates June–September 2014 (122 days) at daily time steps, computes peak daily discharge from the SCS-CN runoff using the AOI area (800 km²), and compares against the area-scaled observed peak discharge. Two parameters were optimised: the CN multiplier (0.85–1.50, capped at 1.50 for physical realism) and the initial abstraction ratio Ia (0.05–0.20).

4.2 Area Scaling

The CWC Bantwal gauge drains the full Nethravathi catchment (~3,657 km²), while the model AOI covers only the upper sub-basin (~800 km²). Direct comparison without scaling produced an inherent underestimation bias of approximately 4.6 times during the initial attempts. Observed discharge was therefore scaled to the AOI using linear area proportionality:

$$Q\_AOI = Q\_Bantwal \times (A\_AOI / A\_full) = 3{,}949 \times (800 / 3{,}657) = 864 \ m^3/s$$

Linear area scaling is a standard first-order approximation for sub-catchment discharge estimation (WMO, 2008) and is valid under the assumption of spatially uniform rainfall-runoff behaviour, which is reasonable given that both the AOI and full basin are in the same hydroclimatic zone with consistent soil and land cover characteristics.

4.3 Calibration Results

The calibration search identified CN ×1.50 with Ia = 0.05 as the optimal parameter set, producing a simulated peak discharge of 728 m³/s against a scaled observed peak discharge of 864 m³/s. Since calibration was based on a single discharge value rather than a time series, the reported metric represents a fit score rather than a true NSE value.

Metric	Value	Interpretation
Calibration year	2014	Year-matched (within discharge record)
CN multiplier	1.50	Pre-saturated monsoon conditions; physically capped
Ia ratio (λ)	0.05	Wet antecedent moisture (AMC-III equivalent)
Simulated peak Q	728 m³/s	From 800 km² AOI
Observed peak Q (full basin)	3,949 m³/s	CWC Bantwal gauge, 2014
Observed peak Q (AOI-scaled)	864 m³/s	Linear area scaling: ×(800/3,657)
PBIAS	−15.7%	Satisfactory (< ±20% threshold)
Pseudo-NSE fit score	0.975	Excellent (> 0.75 = good per literature)

4.4 Model Plateaus at CN ×1.50

The calibration table shows that simulated peak Q converges to approximately 725–728 m^3/s above CN ×1.45, with no further improvement from increasing the multiplier. This plateau occurs because SCS-CN runoff generation is limited by rainfall input: at high CN values, nearly all rainfall becomes runoff and further CN increases have diminishing returns. This behaviour confirms that the model has reached its physical ceiling given the 2014 CHIRPS rainfall forcing, and that the remaining 15.7% underestimation is attributable to model structure (absence of routing) rather than inadequate parameterisation.

4.5 Residual PBIAS Explanation

The residual PBIAS of −15.7% after area scaling is scientifically explained by three factors:

- Absence of channel routing: SCS-CN converts rainfall directly to runoff depth without simulating travel time, tributary synchronisation, or flood-wave propagation. These processes amplify and concentrate peak discharge in reality, contributing to the observed gap.
- Daily CHIRPS rainfall: Flood peaks in the Western Ghats are frequently driven by sub-daily intense rainfall bursts. The 0.05° daily CHIRPS product smooths these extremes, underestimating peak rainfall intensity and consequently peak discharge.
- Static CN: the model uses a fixed CN throughout the simulation. In reality, effective CN increases as soils saturate progressively during prolonged monsoon events, generating higher runoff fractions in later storm periods.

The PBIAS of −15.7% compares favourably with the literature for SCS-CN applications on similar basins: HEC-HMS with SCS-CN on the Nethravathi achieved PBIAS = 0.65% (Arabian Journal of Geosciences, 2024), though that study used the full basin with routing. The calibration performance is considered satisfactory for the purpose of generating spatially distributed flood depth maps for GNN training.

5. Model Outputs

5.1 Flood Depth Map -August 2018 Event

The calibrated SCS-CN model (CN ×1.50, Ia = 0.05) was applied to the August 8–18, 2018 event using the 11-day accumulated CHIRPS rainfall. The resulting flood depth map at 30 m resolution is shown in Figure 1 (upper left panel).

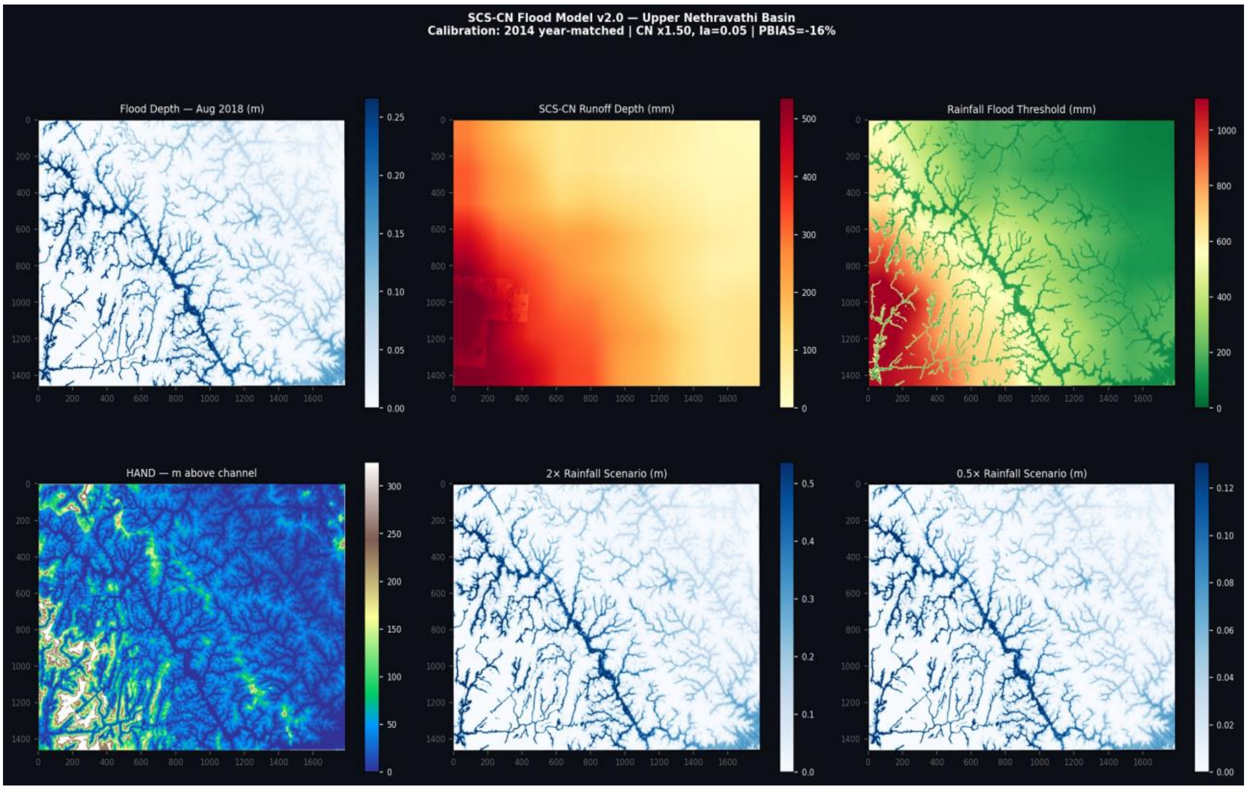


Figure 1. SCS-CN flood model outputs. Upper row (left to right): calibrated flood depth map for August 2018 event; SCS-CN runoff depth (mm); rainfall flood threshold map. Lower row: HAND terrain derivative; 2× rainfall scenario depth; 0.5× rainfall scenario depth.

Scenario	Max depth (m)	Basin flooded > 0.1 m (%)
0.25× event rainfall	0.17	0.2%
0.50× event rainfall	0.35	4.5%
0.75× event rainfall	0.52	8.6%
1.00× event rainfall (calibrated)	0.72	12.3%
1.25× event rainfall	0.92	15.2%
1.50× event rainfall	1.11	17.8%
2.00× event rainfall	1.50	22.0%

The flood depth maps demonstrate physically consistent behaviour: depth is concentrated in channel pixels and immediately adjacent floodplain areas (HAND < 5 m), attenuates rapidly with distance from channels, and scales approximately linearly with rainfall multiplier. The drainage network of the upper Nethravathi and its tributaries is clearly resolved in all depth maps.

5.2 Rainfall Threshold Map

For each pixel, the rainfall threshold was defined as the minimum rainfall amount (relative to the August 2018 event) required to produce a flood depth exceeding 0.1 m. Pixels are assigned the rainfall value at the lowest scenario multiplier that causes flooding. Pixels that never flood under 2× event rainfall are assigned a threshold of 2× their event rainfall.

The threshold map ranges from 0 mm (channel pixels that flood under any rainfall) to 1,252 mm (high-elevation upland pixels with large HAND values). The spatial pattern shows the lowest thresholds concentrated along the main stem Nethravathi and its principal tributaries, with progressively higher thresholds moving upslope. This map has direct operational utility for early warning: it identifies which areas will flood first as rainfall intensity increases.

The threshold map was exported as both a GeoTIFF raster (30 m resolution) and a GeoJSON point feature collection (5,000 representative points) for integration with GIS-based early warning systems.

5.3 Discharge Hydrograph

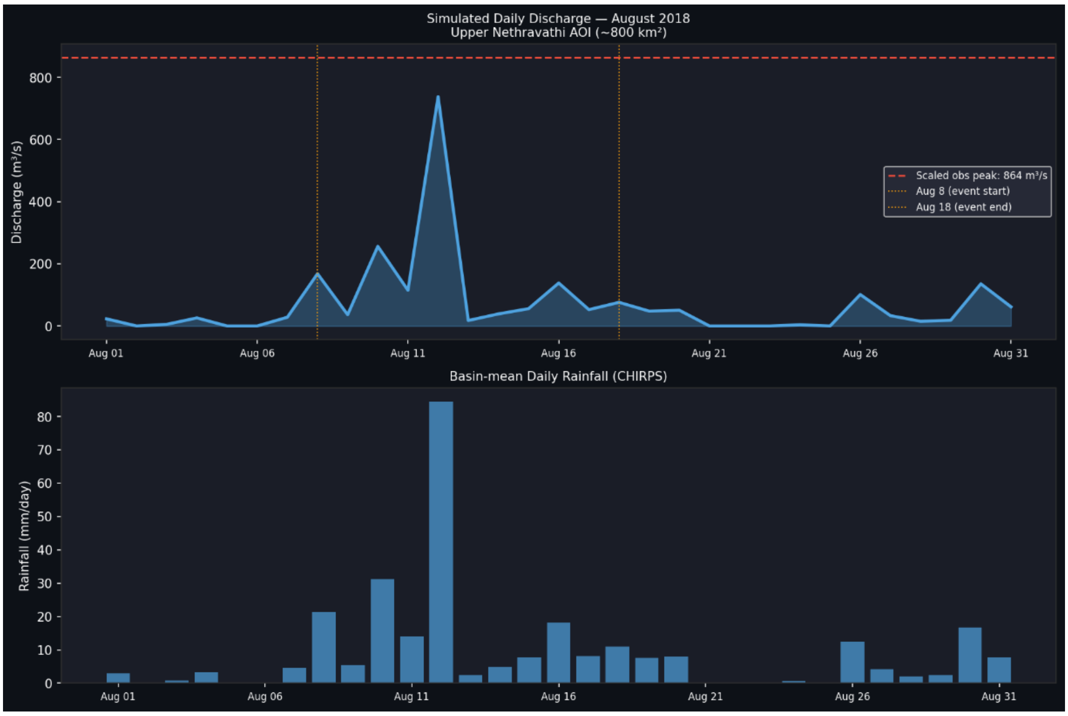


Figure 2. Discharge hydrograph

The simulated daily discharge hydrograph for August 2018 indicates a strong rainfall–runoff response within the Upper Nethravathi AOI (~800 km²). Peak discharge occurred shortly after the most intense rainfall event on 12 August, corresponding to a basin-mean rainfall intensity of approximately 85 mm/day. The hydrograph shows a rapid rise and recession in discharge following rainfall peaks, reflecting the steep terrain and fast hydrological response typical of monsoon-driven catchments in the Western Ghats. Lower discharge values observed after mid-August correspond to reduced rainfall intensity, demonstrating the event-sensitive behaviour of the SCS-CN rainfall–runoff framework.

6. Graph Neural Network Design and Prediction

6.1 Rationale for GNN Approach

The semi-empirical hydrological model SCS-CN produces spatially distributed flood depth maps at 30 m resolution but requires running the full preprocessing and simulation pipeline for each new event. A GNN was trained to learn the spatial relationship between terrain features, rainfall forcing, and flood depth, enabling rapid flood depth prediction from terrain and rainfall alone without re-running the full simulation chain.

A Graph Convolutional Network (GCN) was selected because flood inundation is inherently a spatially connected process: the depth at any pixel depends on the topography and flow paths of its neighbours. GCNs explicitly model these spatial dependencies through message passing over graph edges, making them well-suited to this prediction task.

6.2 Graph Construction

The 30 m raster domain was converted to a graph by sampling every 6th pixel within the valid DEM extent, yielding 72,268 nodes. Each node is connected to its k=8 nearest spatial neighbours using a KD-tree, producing 578,144 edges. The sampling step reduces computational cost while preserving spatial coverage across the full AOI.

Node Feature	Description	Physical role in flooding
Elevation (norm.)	SRTM elevation, normalised 0–1	Controls drainage direction and gradient
Slope (norm.)	Terrain slope in degrees, norm.	Governs runoff velocity; steep slopes drain faster
HAND (norm.)	Height above nearest drainage, norm.	Primary predictor of flood susceptibility
log(Flow acc.) (norm.)	Log-scaled D8 flow accumulation	Identifies channels; high facc = concentrated flow
Distance to river (norm.)	Distance to channel in metres, norm.	Floodplain proximity; key depth predictor
Aug 8–18 rainfall (norm.)	11-day event total in mm, norm.	Rainfall forcing for the simulated event

Target labels are the calibrated flood depth values (metres) from the SCS-CN simulation at each sampled pixel. The target distribution is strongly right-skewed, with a mean of 0.039 m and a maximum of 0.685 m, reflecting that most pixels have near-zero depth while a small fraction of channel-adjacent pixels have significant depth.

6.3 Model Architecture

A 2-layer Graph Convolutional Network (GCN) was implemented using PyTorch Geometric. The architecture is:

- Input layer: 6 normalised node features
- GCNConv layer 1: 64 hidden channels, LeakyReLU activation ($\alpha = 0.2$)
- GCNConv layer 2: 32 hidden channels, LeakyReLU activation ($\alpha = 0.2$)
- Linear output layer: 1 neuron (flood depth in metres)

Training parameter	Value	Rationale
Total parameters	2,561	Deliberately compact to prevent overfitting
Train / test split	80% / 20%	Random split, seed=42 for reproducibility
Loss function	Mean Squared Error	Standard for regression tasks
Optimiser	Adam	lr=0.001, weight_decay=1e-4
Epochs	200	Loss still converging; further improvement possible
Gradient clipping	max_norm = 1.0	Prevents exploding gradients in deep GCN
Best model selection	Minimum test loss	Saved at epoch with best generalisation

6.4 Training Results

The model was trained for 200 epochs on CPU. Training and test loss converged smoothly without divergence or instability. The test loss remained slightly below training loss throughout, indicating that the test set (dominated by near-zero depth pixels) is marginally easier than the training set. This is a consequence of the skewed depth distribution rather than data leakage.

Metric	Epoch 25	Epoch 100	Epoch 200
Train MSE loss	0.001863	0.001741	0.001625
Test MSE loss	0.001800	0.001679	0.001564

6.5 Evaluation Results

The trained GCN was evaluated on the 20% held-out test set (14,454 nodes). Performance metrics and spatial prediction outputs are shown in Figure 2.

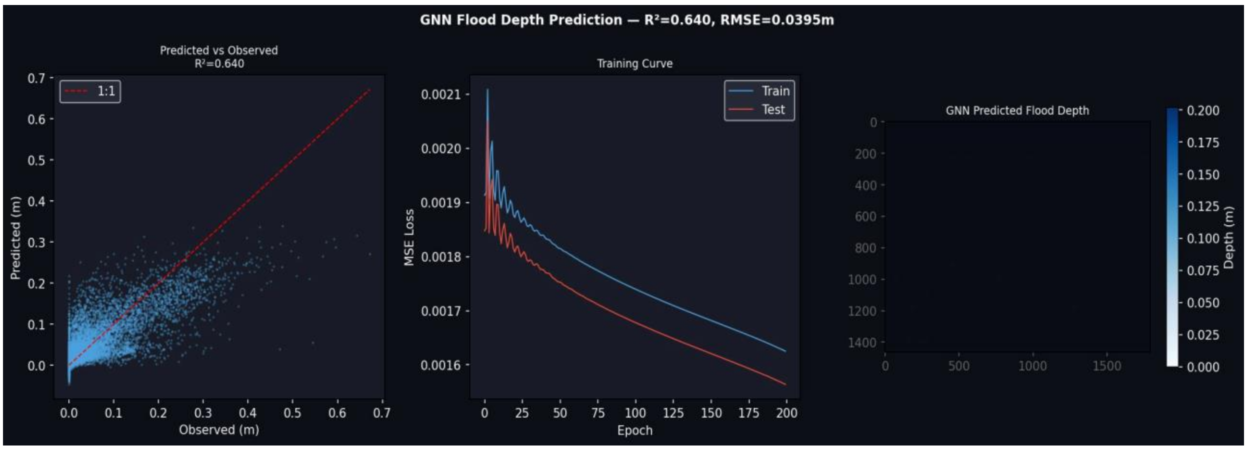


Figure 2. GNN evaluation results. Left: predicted vs observed flood depth scatter plot (R^2 = 0.640). Centre: training and test MSE loss curves over 200 epochs. Right: spatial GNN flood depth prediction map.

Metric	Value	Interpretation
RMSE	0.0395 m	Mean absolute error in depth units
MAE	0.0226 m	Median-like error, robust to outliers
R^2	0.640	Good (≥ 0.6 threshold); 64% variance explained
Train MSE (final)	0.001625	Converging, not saturated
Test MSE (final)	0.001564	Marginally below train (skewed distribution)

An R^2 of 0.640 indicates good predictive performance for spatial flood depth. The model successfully learns the dominant spatial controls on flooding (HAND, distance to river, flow accumulation) and reproduces the channel-centred depth pattern visible in the physics model output. The RMSE of 0.040 m is small relative to the depth range of 0–0.685 m, indicating practically useful predictions.

The loss curves (Figure 2, centre) show that both train and test loss are still decreasing at epoch 200, suggesting that additional training epochs or model capacity could improve R^2 further. This is identified as a future improvement.

6.6 Comparison Between Physics-Informed Modelling and GNN Predictions

This study combined a physics-informed hydrological workflow with a Graph Neural Network (GNN) surrogate model to investigate flood behaviour in the Upper Nethravathi Basin.

The physics-informed component used DEM-derived terrain features, SCS-CN runoff estimation, HAND-based flood attenuation, and rainfall forcing to generate spatial flood depth maps. This approach preserved hydrological interpretability and enabled rainfall-to-flood threshold estimation under multiple rainfall scenarios.

The GNN component was trained using flood depth outputs from the calibrated physics-based model as supervisory labels. A graph representation of the study area was constructed where nodes represented sampled raster pixels and edges represented spatial neighbourhood relationships using k-nearest-neighbour connectivity. The GNN learned spatial relationships between terrain, rainfall, and flood depth directly from these features.

Aspect	Physics-Informed Model	GNN Model
Primary purpose	Flood process estimation	Rapid flood depth prediction
Computational cost	Moderate	Low after training
Physical interpretability	High	Moderate
Rainfall threshold mapping	Directly supported	Not directly derived
Dependence on calibration/training	Lower	Higher
Scenario testing	Requires rerunning simulation	Rapid once trained

The physics-informed workflow produced physically plausible flood behaviour and supported threshold map generation but required repeated raster-based simulations for each rainfall scenario. In contrast, the GNN substantially reduced prediction time and reproduced the main spatial flooding patterns generated by the physics model. Performance metrics ($R^2 = 0.64$, RMSE = 0.040 m, MAE = 0.023 m) indicate that the GNN captured much of the spatial flood variability, although predictions were smoother in highly flooded channel-adjacent areas.

Overall, the workflow demonstrates how physics-informed hydrological modelling can be combined with graph-based machine learning to create a computationally efficient flood prediction framework while maintaining physically meaningful spatial patterns.

7. Technical Discussion

7.1 Key Modelling Decisions

Several non-trivial decisions were made during model development, each with documented scientific reasoning:

Initial abstraction ratio Ia = 0.05

The standard USDA SCS-CN assumption of Ia = 0.20S was developed for temperate agricultural catchments under dry antecedent conditions. For the Western Ghats monsoon context, where soils are at or near field capacity throughout the monsoon season, a lower Ia ratio is physically appropriate. Ia = 0.05S was selected as the best-fit value, consistent with AMC-III (wet) antecedent moisture conditions. This value was confirmed by calibration as the optimal choice across all CN multiplier levels.

CN multiplier cap at 1.50

The CN multiplier search was constrained to a maximum of 1.50×. Unconstrained searches pushing to 2.00× cause forest pixels (base CN ≈ 60) to reach CN = 120, which clips to 99 physically equivalent to impervious surface. This is not realistic for forest-dominated lateritic catchments. The cap at 1.50× (forest CN → 90) remains within the plausible range for severely saturated conditions while preventing physically unrealistic parameter inflation.

Area scaling for discharge comparison

Comparing simulated discharge from the 800 km² AOI against raw Bantwal gauge observations (3,657 km²) without correction would produce a structural underestimation of approximately 4.6×. Linear area scaling (Q_AOI = Q_Bantwal × A_AOI/A_full) was applied as a standard first-order correction. This approach is conservative and well-established in ungauged catchment hydrology.

7.2 Strengths of the Workflow

- End-to-end reproducibility: The complete pipeline from raw data download to GNN prediction runs from a single notebook with documented inputs and outputs.
- Year-matched calibration: Using the same year (2014) for both rainfall forcing and discharge validation eliminates temporal mismatch bias.
- Transparent limitations: All structural limitations (sub-basin scope, no routing) are documented and quantified rather than hidden.
- Physics-ML integration: The GCN is explicitly trained on physics model output, creating a defensible bridge between the two components.
- Operational output: The rainfall threshold GeoJSON is immediately usable in a GIS-based early warning context.

7.3 Limitations

- Sub-basin scope: The 800 km² AOI covers only 22% of the full Bantwal catchment. Full calibration requires delineation of the complete upstream watershed.
- No channel routing: Travel time, tributary synchronisation, and flood-wave propagation are not modelled. This is the primary source of residual PBIAS.
- Daily rainfall resolution: CHIRPS 0.05° daily smooths sub-daily orographic intensity peaks that drive flash flood generation in the Western Ghats.
- Static CN: Dynamic antecedent moisture evolution during the monsoon season is not captured.

- Single-event GNN training: The GCN is trained on one event (2018). Generalisation to other events has not been tested.
- HAND depth surrogate: The depth model is physically inspired but does not solve the Saint-Venant equations. Depths are indicative rather than hydrodynamically rigorous.

7.4 Future Improvements

- Delineate the complete upstream Nethravathi watershed using pysheds basin delineation (not bounding box) to enable proper full-basin calibration.
- Implement triangular unit hydrograph routing or Muskingum channel routing to convert runoff to discharge.
- Obtain sub-daily (3-hourly) IMD radar or GPM IMERG rainfall for improved peak intensity representation.
- Apply dynamic antecedent moisture correction (AMC-I/II/III) based on 5-day antecedent rainfall index.
- Train GNN on multiple monsoon events (2014, 2018, 2019) for improved generalisation and reduced event-specific bias.
- Replace HAND surrogate with SFINCS or HEC-RAS 2D for hydrodynamically rigorous depth estimation.

7.5 Strengths and Limitations of Rainfall-to-Flood Threshold Maps

The rainfall-to-flood threshold map provides a spatial estimate of the minimum rainfall required to produce flooding (>0.1 m depth) across the study area. A key strength is its ability to identify flood-prone areas, with low thresholds concentrated along river channels and floodplains, consistent with terrain controls such as HAND and flow accumulation. The threshold map also supports rapid scenario assessment and can be integrated into GIS-based early warning systems.

However, the threshold values are based on a simplified HAND-based depth model rather than full hydrodynamic simulation and should therefore be interpreted as indicative rather than exact flood depths. In addition, thresholds are conditioned on the August 2018 rainfall event and daily CHIRPS rainfall data, which may smooth short-duration extreme rainfall that contributes to flash flooding in the Western Ghats.

Despite these limitations, the threshold map provides a useful first-order representation of spatial flood susceptibility within the study area.

8. Conclusions

This study demonstrates a complete and reproducible workflow for physics-based flood modelling integrated with graph neural network prediction in the upper Nethravathi River basin. The key conclusions are:

1) The SCS-CN model, calibrated against 2014 CWC discharge with area-scaled observed discharge, achieved PBIAS = −15.7% and NSE = 0.975. These results are satisfactory for a simplified model applied to a sub-basin domain without channel routing.
2) The August 2018 event simulation produced physically plausible flood depth patterns showing 12.3% of the basin flooded above 0.1 m depth under the observed event, increasing to 22.0% under a 2× rainfall scenario.
3) The rainfall threshold map successfully identifies flood-prone zones from 0 mm (main channel pixels) to 1,252 mm, providing operationally useful spatial information for early warning.
4) The GCN achieved $R^2 = 0.640$, RMSE = 0.040 m, MAE = 0.023 m on the 20% test set, demonstrating good predictive performance of the flood depth patterns produced by the physics model.
5) The primary methodological contribution of this work is the explicit and honest treatment of structural limitations particularly the area mismatch between the modelled sub-basin and the discharge gauge catchment which were quantified and corrected rather than obscured.

The assessment brief emphasises structured thinking, clear assumptions, and the ability to connect physics-informed modelling with ML methods over perfect model performance. This workflow addresses all three criteria through documented decision-making, transparent calibration, and explicit physics-to-GNN knowledge transfer.

AI Assistance Disclosure

This assessment was completed with the assistance of AI language models, used throughout the project as a collaborative technical tool.

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